

DRAWING LIST FOR TEST PACKAGE

PAC-509

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	509	4FA	OD90-PAC-HP2-N-6513-01	1 OF 2	1	CJ-M15W14-PR-A20-00151	1A	
2	509	4FA	OD50-PAC-HP2-N-6509-01	1 OF 1	1	CJ-M15W14-PR-A20-00151	1A	
3	509	4FA	OD32-PAC-HP2-N-6524-01	1 OF 2	1	CJ-M15W14-PR-A20-00151	1A	
4	509	4FA	OD32-PAC-HP2-N-6524-02	2 OF 2	1	CJ-M15W14-PR-A20-00151	1A	
5	509	4FA	OD50-PAC-HP2-N-6510-01	1 OF 1	1	CJ-M15W14-PR-A20-00151	1A	
6	509	4FA	OD50-PAC-HP2-N-6511-01	1 OF 1	0	CJ-M15W14-PR-A20-00151	1A	
7	509	4FA	OD90-PAC-HP2-N-6513-02	2 OF 2	0	CJ-M15W14-PR-A20-00151	1A	
8	509	5FA	OD90-PAC-HP2-N-6513-01	1 OF 1	0	CJ-M15W14-PR-A20-00151	1A	
9	509	RFA	OD90-PAC-HP2-E-6513-01	1 OF 4	1	CJ-M15W14-PR-A20-00151	1A	
10	509	RFA	OD90-PAC-HP2-E-6513-02	2 OF 4	1	CJ-M15W14-PR-A20-00151	1A	
11	509	RFA	OD32-PAC-HP2-E-6517-01	1 OF 2	1	CJ-M15W14-PR-A20-000105/151	1A	
12	509	RFA	OD32-PAC-HP2-E-6517-02	2 OF 2	1	CJ-M15W14-PR-A20-000105/151	1A	
13	509	RFA	OD32-PAC-HP2-E-6558-01	1 OF 1	1	CJ-M15W14-PR-A20-000105	1A	
14	509	RFA	OD90-PAC-HP2-E-6513-03	3 OF 4	1	CJ-M15W14-PR-A20-00151	1A	
15	509	RFA	OD32-PAC-HP2-E-6516-01	1 OF 2	1	CJ-M15W14-PR-A20-000104/151	1A	
16	509	RFA	OD32-PAC-HP2-E-6516-02	2 OF 2	1	CJ-M15W14-PR-A20-000104151	1A	
17	509	RFA	OD32-PAC-HP2-E-6556-01	1 OF 1	0	CJ-M15W14-PR-A20-000104	1A	
18	509	RFA	OD32-PAC-HP2-E-6514-01	1 OF 2	1	CJ-M15W14-PR-A20-000102/151	1A	
19	509	RFA	OD32-PAC-HP2-E-6514-02	2 OF 2	1	CJ-M15W14-PR-A20-000102/151	1A	
20	509	RFA	OD32-PAC-HP2-E-6552-01	1 OF 1	1	CJ-M15W14-PR-A20-000102	1A	

DRAWING LIST FOR TEST PACKAGE

PAC-509

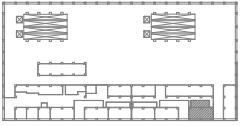
PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
21	509	RFA	OD32-PAC-HP2-E-6515-01	1 OF 2	1	CJ-M15W14-PR-A20-000151	1A	
22	509	RFA	OD32-PAC-HP2-E-6515-02	2 OF 2	1	CJ-M15W14-PR-A20-000151	1A	
23	509	RFA	OD32-PAC-HP2-E-6554-01	1 OF 1	1	CJ-M15W14-PR-A20-000103	1A	
24	509	RFA	OD32-PAC-HP2-E-6523-01	1 OF 2	1	CJ-M15W14-PR-A20-000110/151	1A	
25	509	RFA	OD32-PAC-HP2-E-6523-02	2 OF 2	1	CJ-M15W14-PR-A20-000110/151	1A	
26	509	RFA	OD32-PAC-HP2-E-6566-01	1 OF 1	1	CJ-M15W14-PR-A20-000110	1A	
27	509	RFA	OD90-PAC-HP2-E-6513-04	4 OF 4	1	CJ-M15W14-PR-A20-00151	1A	
28	509	RFA	OD32-PAC-HP2-E-6522-01	1 OF 2	1	CJ-M15W14-PR-A20-000109/151	1A	
29	509	RFA	OD32-PAC-HP2-E-6522-02	2 OF 2	1	CJ-M15W14-PR-A20-000109/151	1A	
30	509	RFA	OD32-PAC-HP2-E-6564-01	1 OF 1	1	CJ-M15W14-PR-A20-000109	1A	
31	509	RFA	OD32-PAC-HP2-E-6520-01	1 OF 2	1	CJ-M15W14-PR-A20-000107/151	1A	
32	509	RFA	OD32-PAC-HP2-E-6520-02	2 OF 2	1	CJ-M15W14-PR-A20-000107/151	1A	
33	509	RFA	OD32-PAC-HP2-E-6560-01	1 OF 1	1	CJ-M15W14-PR-A20-000107	1A	
34	509	RFA	OD32-PAC-HP2-E-6521-01	1 OF 2	1	CJ-M15W14-PR-A20-000108/151	1A	
35	509	RFA	OD32-PAC-HP2-E-6521-02	2 OF 2	1	CJ-M15W14-PR-A20-000108/151	1A	
36	509	RFA	OD32-PAC-HP2-E-6562-01	1 OF 1	1	CJ-M15W14-PR-A20-000108	1A	

ITEM NO.	W14-T-PAC-4605 A
SERVICE	PAC STORAGE TANK
Q'TY	1
TYPE	VERTICAL & CYLINDRICAL
SIZE	4.0 x 4.5 (EH/4.2) 1A
CAPA.	m3 57 (EH/53)
MAT'L	FRP

대외비

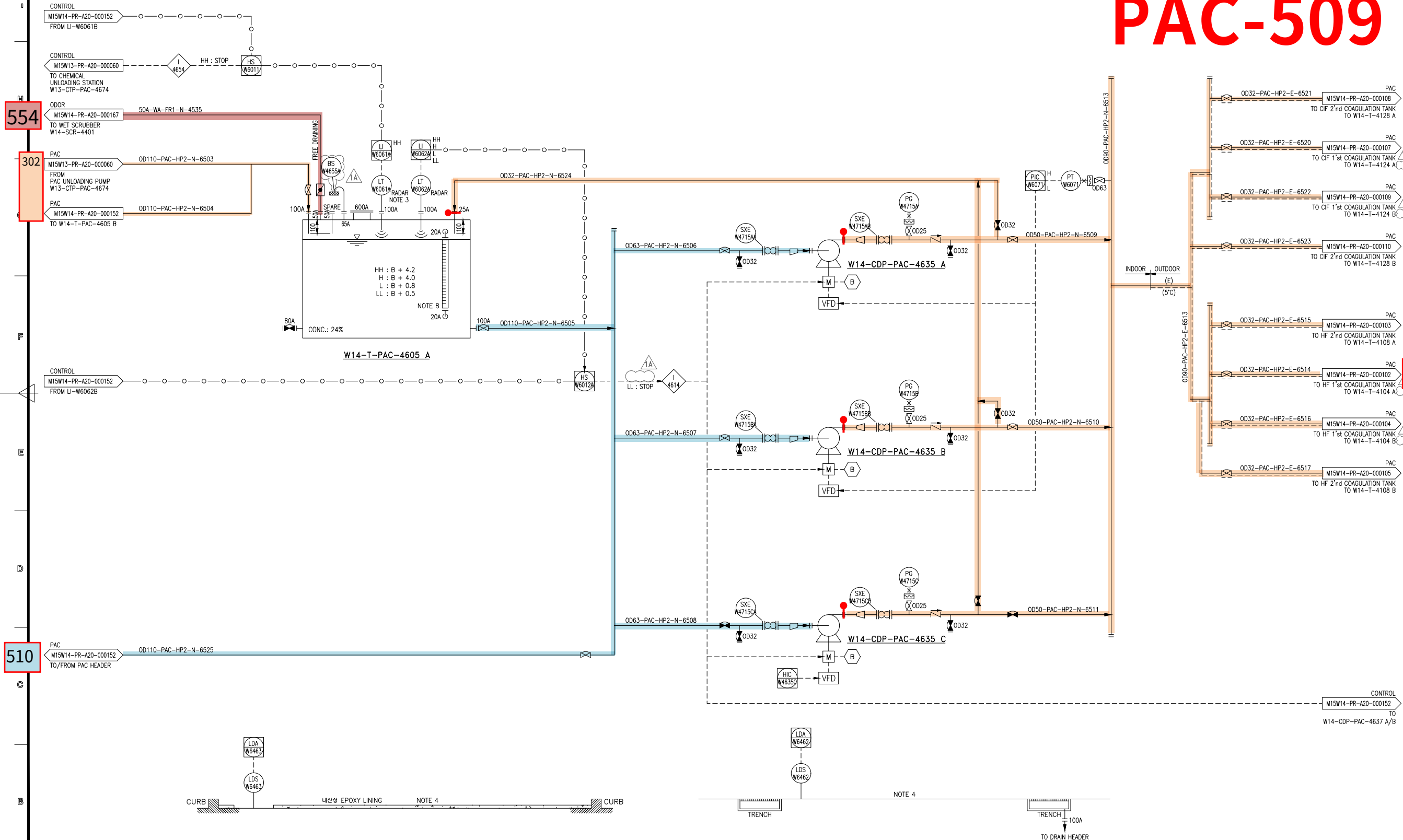


KEY PLAN (4F)

NOTE

- CENTRIFUGAL PUMP
MAT'L 표기 : CASING/IMPELLER
- 약품종류에 따라 LINE 색상을 구분하여야 한다.
- CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
- CHEMICAL STORAGE TANK CURB 및 PUMP AREA에서
되는 DRAIN은 인근의 TRENCH로 접속되어,
RAIN HEADER로 이동된다.
- INTERLOCK NO. I-4654
(W13-CTP-PAC-4674 AUTO STOP)
REMOTE STATUS에서 W14-T-PAC-4605 A/B
LEVEL (U-W6061 A/B) HIGH HIGH STOP.
- INTERLOCK NO. I-4614
(W14-CDP-PAC-4635 A/B/C
REMOTE STATUS에서 W14-T-PAC-4605 A/B
LEVEL (U-W6062 A/B) LOW LOW STOP.
시 CLOSE 되는 배관에는 LOTO 설치.
TANK의 역변 레벨 개시지는 VENDOR에서 공급한다.

PAC-509



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1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	실명 DESCRIPTION	날짜 DATE

제도	이인학
검토	한진희
승인	박동명
제작사	SK 에코플랜트

공사명
PROJECT TITLE
M15X Ph4 WWT Project

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
PAC STORAGE TANK & FEED PUMP
(1/2)

속척
SCALE
NONE

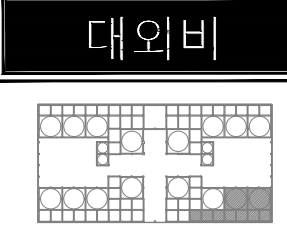
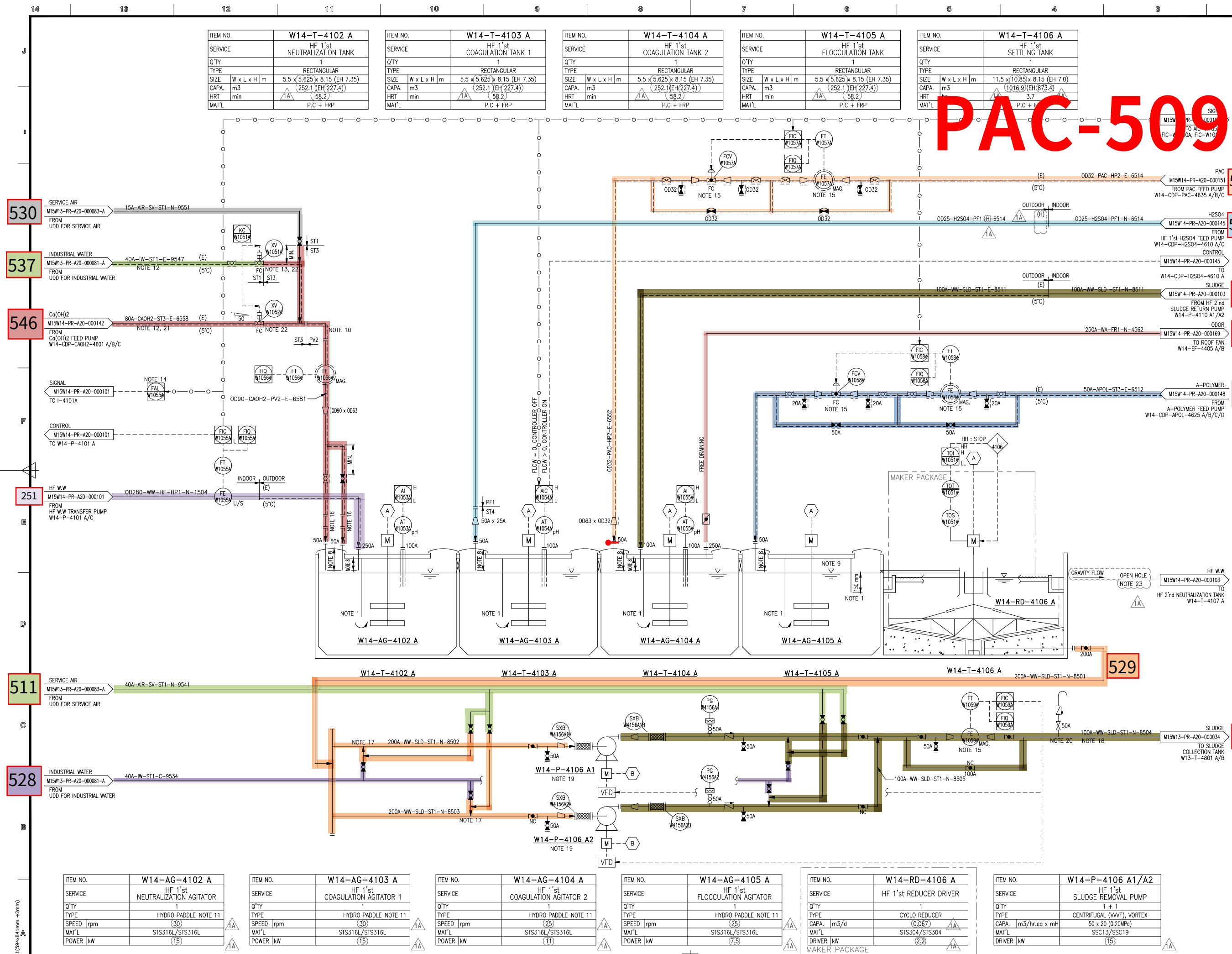
날짜
DATE
2025. 08. 14

PROJECT NO. 22330D

도면번호
SHEET NO. CJ-M15W14-PR-A20-000151

REV. 1A

ITEM NO.	W14-CDP-PAC-4635 A/B/C
SERVICE	PAC FEED PUMP
Q'TY	2 + 1
TYPE	CENTRIFUGAL (VWF)
CAPA.	m3/hr.ea x MPa 4 x 0.46
MAT'L	FRP/FRP
POWER kW	11 1A



KEY PLAN (5F)

NOTE

1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브
설치할 것.
2. 3.2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는
PLATFORM 구성할 것.
3. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
사전 시 CLOSE 되는 배관에는 LOTO 설치.
4. 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
5. 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
6. 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록
최소 길이로 변경 할 것.
AB 으로부터 100mm 떨어질 것
7. 10. Co(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것
SS304 + FRP LINING 적용.
8. 11. VENDOR에서 결정한다.
9. 12. 라인은 상부에서 시수 도입하도록 설치한다.
W1051A OPEN (12'간마다 15') 일 경우 :
XV-W1052A CLOSE
10. 14. W14-P-4101 A가 RUN 후 30초 후부터 동작한다.
11. 15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및
CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
12. 16. 구간 투명 PVC 적용. (주입 여부 확인용)
13. 17. SHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
14. 18. SLUDGE REMOVAL PUMP 토출 배관은 PUMP SUCTION 속의
TANK 수위보다 HIGH POINT 형성.
15. 19. SLUDGE PUMP는 TIMER에 의해 운전함.
16. 20. HIGH POINT에 SIPHON BREAKER 설치함.
(CHECK VALVE의 경우 DUAL PLATE TYPE로 적용)
17. 21. 소액의 공금 HEADER에서 주입되는 POINT 별
ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE,
SPEC. BREAK 까지 MIN. DISTANCE 적용하여, 위험 최소화
및 주입배관 고제를 용이하게 한다.
18. 22. XVM SPEED CONTROLLER (ACCESSORY) 설치.
19. 23. CONCRETE WALL OPENING을 통한 유체 이송
OPENING SIZE IS 800mm x 400mm

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE

제	도	이	인	학
DRAWN BY				
검	토	한	진	회
CHECKED BY				
인	박	동	영	
REVISED BY				
제	작	사		

SK 에코플랜트

공 사 명

PROJECT TITLE

M15X Ph4 WWT Project

도면명

DWG. TITLE

PIPING & INSTRUMENT DIAGRAM

HF 1st COAGULATION & SETTLING

TANK A (1/2)

속

SCALE

NONE

날

DATE

2025. 08. 14

PROJECT NO.

22330D

도면번호

SHEET NO.

CJ-M15W14-PR-A20-000102

REV.

1A

ITEM NO.	W14-T-4107 A
SERVICE	HF 2'nd NEUTRALIZATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 5.5 x 5.625 x 8.15 (EH 6.5)
CAPA.	m3 (252.1 (EH 201.1))
HRT	min (51.5)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4108 A
SERVICE	HF 2'nd COAGULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 5.5 x 5.625 x 8.15 (EH 6.5)
CAPA.	m3 (252.1 (EH 201.1))
HRT	min (51.5)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4109 A
SERVICE	HF 2'nd FLOCCULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 5.7 x 5.625 x 8.15 (EH 6.5)
CAPA.	m3 (261.3 (EH 208.4))
HRT	min (53.4)
MAT'L	P.C + FRP

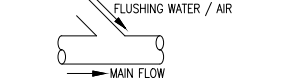
ITEM NO.	W14-T-4110 A
SERVICE	HF 2'nd SETTLING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 11.7 x 10.85 x 8.15 (EH 6.5)
CAPA.	m3 (1034.6 (EH 825.1))
HRT	hr 3.5
MAT'L	P.C + FRP

ITEM NO.	W14-LP-4110 A
SERVICE	LAMELLAR PLATE
Q'TY	1
CAPA.	W x L x H m BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 6,450m3/d)
MAT'L	ABS+SS304

ITEM NO.	W14-LP-4110 A
SERVICE	LAMELLAR PLATE
Q'TY	1
CAPA.	W x L x H m BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 6,450m3/d)
MAT'L	ABS+SS304

KEY PLAN (5F)

- NOTE
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
 2. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
 5. 온천 시 CLOSE 되는 배관에는 LOTO 설치.
 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 변경 할 것.
(SLAB으로부터 100mm 떨어질 것)
 9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 10. 위에 거주층의 상계사창은 추후 VENDOR에서 결정한다.
 11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.
 12. FLUSHING 용 WATER / AIR 배관은 45도 서면으로 연결할 것.

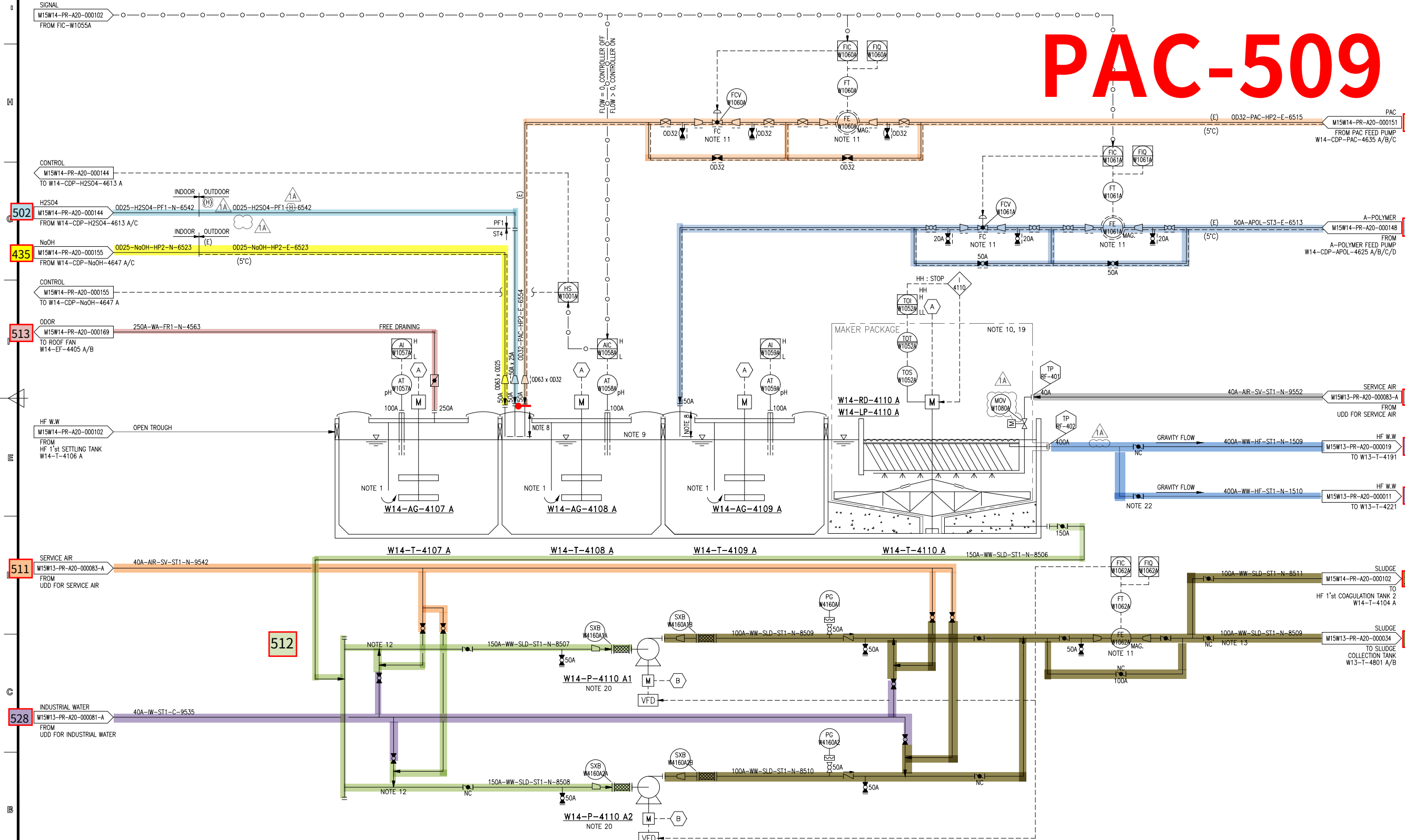


13. SLUDGE REMOVAL PUMP 토출 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 향상.
14. DELETED.
15. DELETED.
16. DELETED.
17. DELETED.
18. DELETED.
19. 검사한 장소 및 유지관리시설을 PACKAGE VENDOR 공급범위에 포함할 것.
20. SLUDGE PUMP는 TIMER에 의해 운전됨.
21. DELETED.
22. VALVE 이용이 접근 가능해야 하며, LIMIT SWITCH가 육안에 보이기 설치 해야 함.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
REV.	실명	날짜
DRAWN BY	이인학	
CHECKED BY	한진희	
APPROVED BY	박동명	
제작사	SK 에코플랜트	

공사명	PROJECT TITLE
M15X Ph4 WWT Project	
도면명	DWG. TITLE
PIPING & INSTRUMENT DIAGRAM	
HF 2'nd COAGULATION & SETTLING TANK A (1/2)	
축척	SCALE
NONE	
날짜	DATE
2025. 08. 14	
PROJECT NO.	22330D
도면번호	SHEET NO.
CJ-M15W14-PR-A20-000103	
REV.	1A

PAC-509



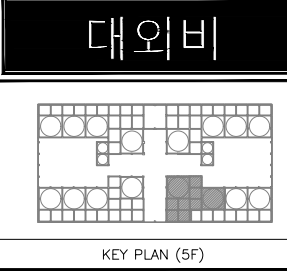
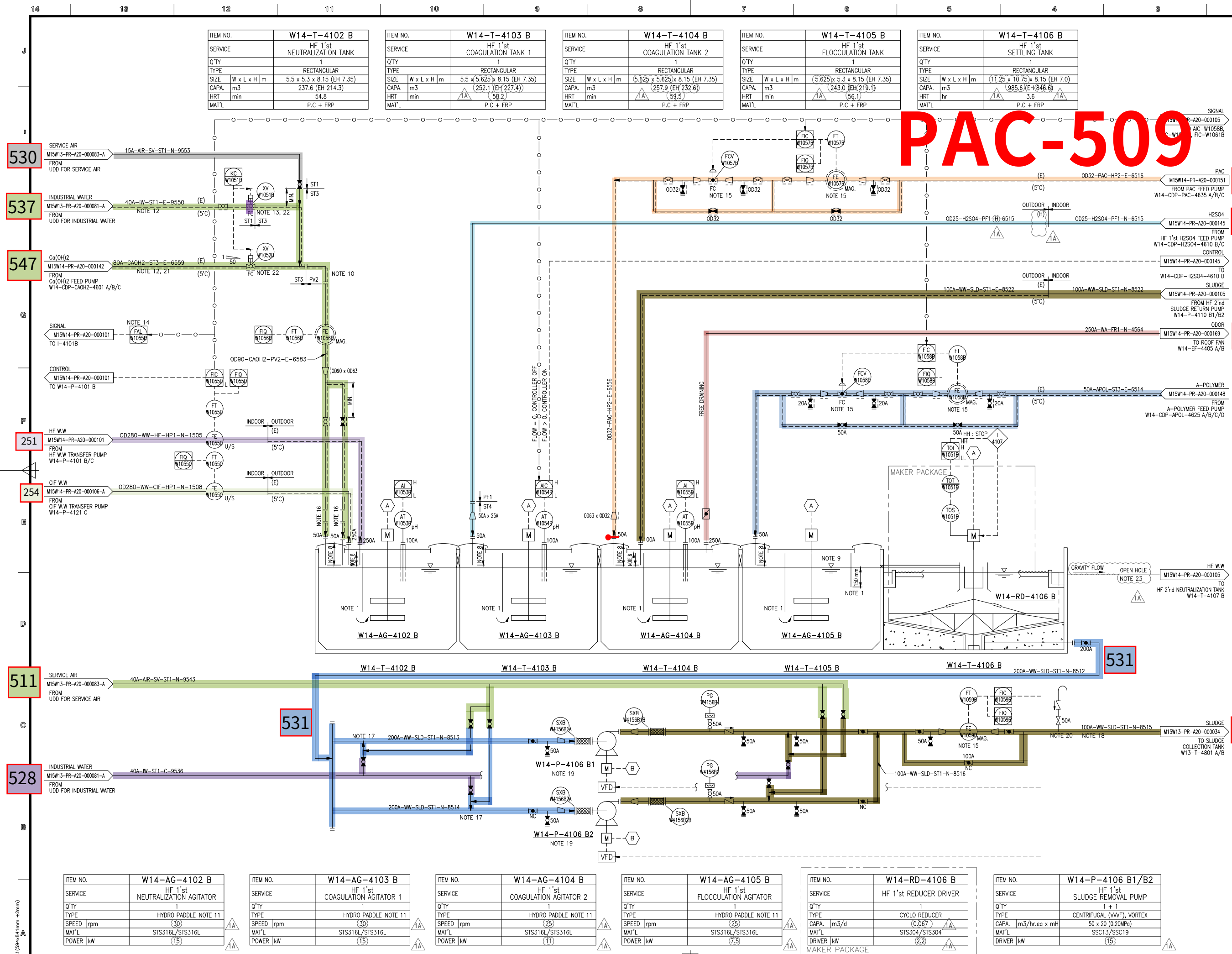
ITEM NO.	W14-AG-4107 A
SERVICE	HF 2'nd NEUTRALIZATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (30)
MAT'L	STS316L/STS316L
POWER	kw (15)

ITEM NO.	W14-AG-4108 A
SERVICE	HF 2'nd COAGULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kw (11)

ITEM NO.	W14-AG-4109 A
SERVICE	HF 2'nd FLOCCULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kw (7.5)

ITEM NO.	W14-RD-4110 A
SERVICE	HF 2'nd REDUCER DRIVER
Q'TY	1
TYPE	CYCLO REDUCER
CAPA.	m3/d (0.067)
MAT'L	STS304/STS304
POWER	kw (2.2)

ITEM NO.	W14-P-4110 A1/A2
SERVICE	HF 2'nd SLUDGE RETURN PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VVF), VORTEX
CAPA.	m3/hr.ea x mH 50 x 30 (0.29MPa)
MAT'L	SSC13/SSC19
POWER	kw (30)



KEY PLAN (5F)

- NOTE**
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
 2. NK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 1개 이상.
 3. 2.2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. INHOLE : 2개 이상 설치한다.
INHOLE COVER : FRP
시 CLOSE 되는 배관에는 LOTO 설치.
 5. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 6. AGITATOR MAT'L 표기 : SHAFT/BLADE
 7. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 변경 할 것.
B 으로부터 100mm 떨어질 것
 8. FRP LINING TANK의 SLEEVE는 504 + FRP LINING 적용.
 9. 10. Co(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것
 11. 주용 VENDOR에서 결정한다.
 12. 라인은 상부에서 시공 시 유입하드록 설치한다.
 13. W1051B OPEN (12시간마다 15초) 일 경우 : W1052B CLOSE
 14. W14-P-4101 B가 RUN 후 30초 후부터 동작한다.
 15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
 16. 구간 투명 PVC 적용. (주입 여부 확인용)
 17. SHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
-

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30

수정	REV.	실명	DESCRIPTION	날짜	DATE
도	DRAWN BY	이인학			
검	CHECKED BY	한진희			
인	APPROVED BY	박동명			

SK 에코플랜트

공 사 명
PROJECT TITLE
M15X Ph4 WWT Project

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
HF 1st COAGULATION & SETTLING
TANK B (2/2)

속 척
SCALE
NONE

날 짜
DATE
2025. 08. 14

PROJECT NO.	22330D	REV.	
도면번호	CJ-M15W14-PR-A20-000104	1A	

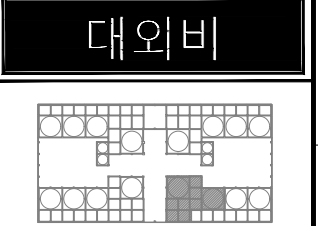
ITEM NO.	W14-T-4107 B
SERVICE	HF 2'nd NEUTRALIZATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (5.75 x 5.45) x 8.15 (EH 6.5)
CAPA.	m3 (255.4 (EH 203.7)
HRT	min (52.1)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4108 B
SERVICE	HF 2'nd COAGULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (5.75 x 5.175) x 8.15 (EH 6.5)
CAPA.	m3 (242.5 (EH 193.4)
HRT	min (49.5)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4109 B
SERVICE	HF 2'nd FLOCCULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (5.625 x 5.175) x 8.15 (EH 6.5)
CAPA.	m3 (237.2 (EH 189.2)
HRT	min (48.4)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4110 B
SERVICE	HF 2'nd SETTLING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (11.375 x 10.85) x 8.15 (EH 6.5)
CAPA.	m3 (1005.9 (EH 802.2)
HRT	hr (3.4)
MAT'L	P.C + FRP

ITEM NO.	W14-LP-4110 B
SERVICE	LAMELLAR PLATE
Q'TY	1
CAPA.	W x L x H m BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 6,450m3/d)
MAT'L	ABS+SS304



KEY PLAN (5F)

- NOTE
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m 이상 AIR HOLE 설치.
 2. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. MANHOLE : 2개 이상 설치한다.
HOLE COVER : FRP

5. 시 CLOSE 되는 배관에는 LOTO 설치.
6. TRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
7. AGITATOR MAT'L 표기 : SHAFT/BLADE
8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 변경 할 것.
(SLAB 으로부터 100mm 떨어질 것)
9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.

10. 특수용의 설계사항은 추후 VENDOR에서 결정한다.
11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
12. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.

13. SLUDGE REMOVAL PUMP 토출 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 향상.
14. DELETED.
15. DELETED.
16. DELETED.
17. DELETED.
18. DELETED.

19. 청소 및 유지관리시설을 PACKAGE VENDOR 공급받아야 함.
20. SLUDGE PUMP는 TIMER에 의해 운전됨.
21. DELETED.
22. 용이 접근 가능해야 하며, LATCH가 육안에 보이게 설치 해야 함.

23. DELETED.
24. DELETED.
25. DELETED.
26. DELETED.
27. DELETED.
28. DELETED.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
375	APPROVAL FOR CONSTRUCTION	24.09.30
375	실명 DESCRIPTION	날짜 DATE

제 도	이 인 학
DRAWN BY	
검 토	한 진 희
CHECKED BY	
승 인	박 동 명
APPROVED BY	
제 작 사	

SK 에코플랜트

공 사 명
PROJECT TITLE
M15X Ph4 WWT Project

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
HF 2'nd COAGULATION & SETTLING
TANK B (2/2)

속 척
SCALE
NONE

날 짜
DATE
2025. 08. 14

PROJECT NO. 22330D REV. 1A
도면번호
SHEET NO. CJ-M15W14-PR-A20-000105

PAC-509

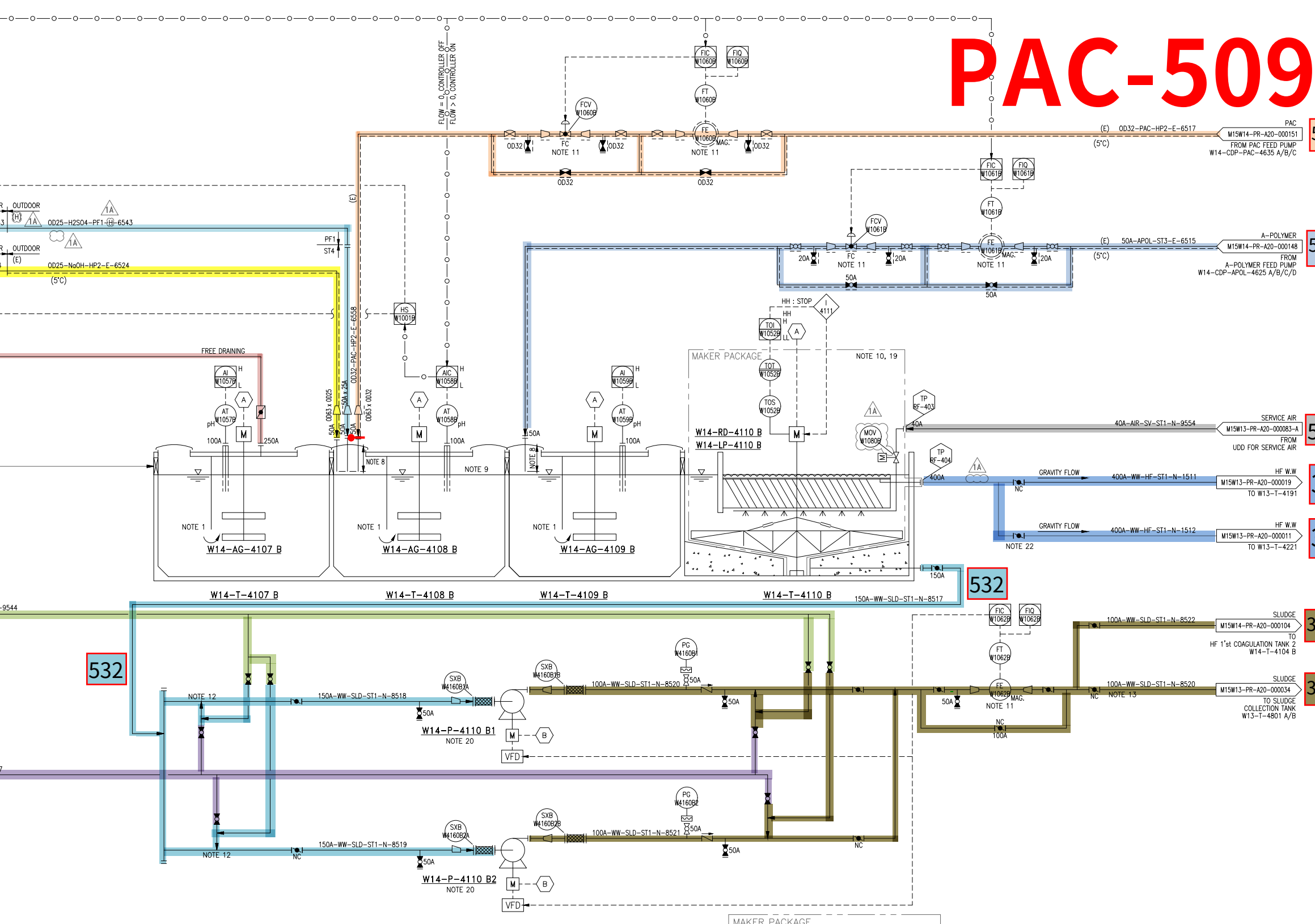
ITEM NO.	W14-AG-4107 B
SERVICE	HF 2'nd NEUTRALIZATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (30)
MAT'L	STS316L/STS316L
POWER	kw (15)

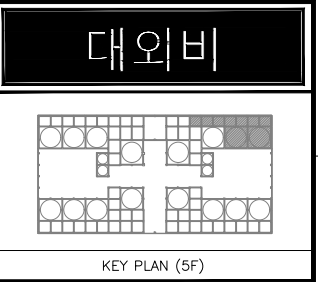
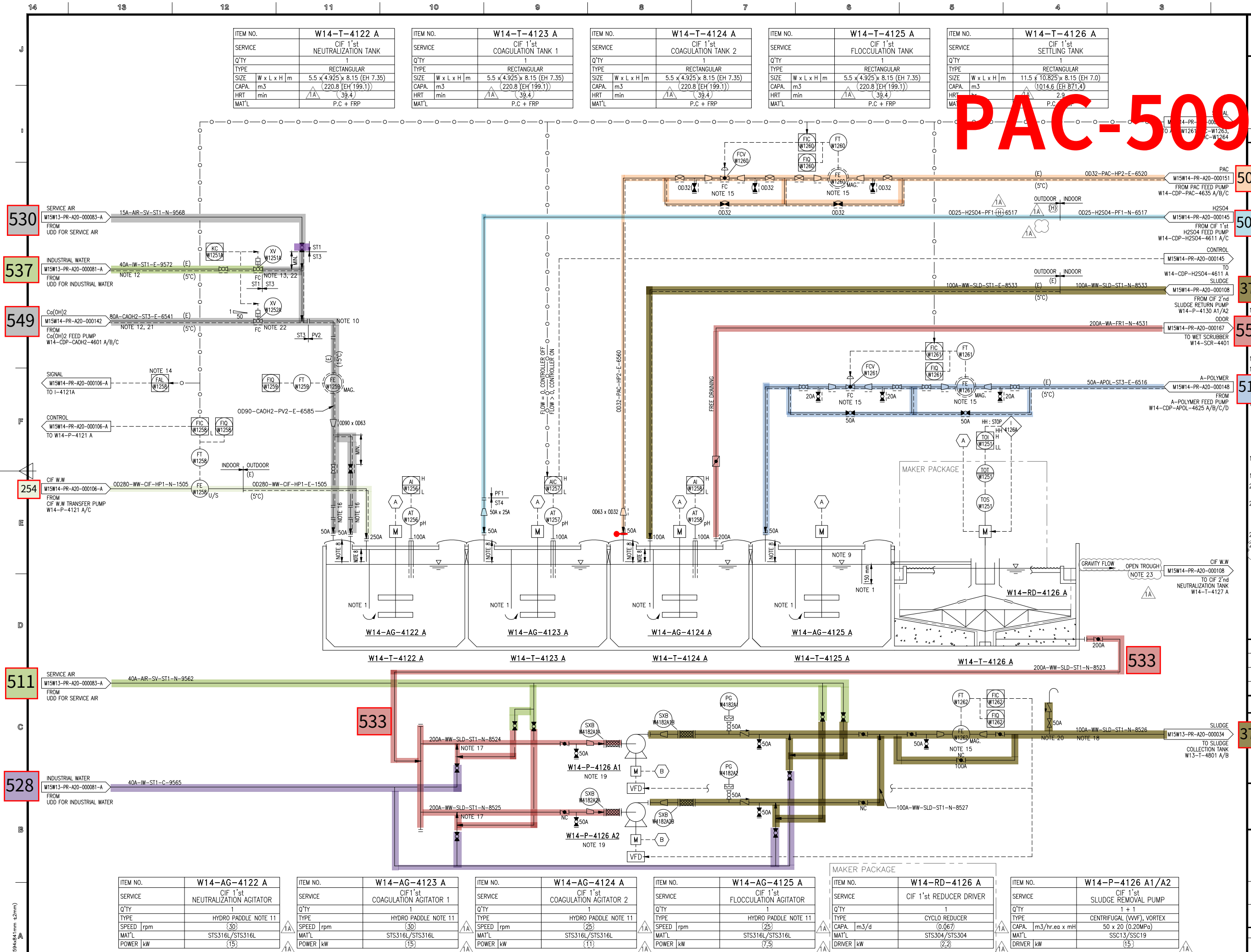
ITEM NO.	W14-AG-4108 B
SERVICE	HF 2'nd COAGULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kw (15)

ITEM NO.	W14-AG-4109 B
SERVICE	HF 2'nd FLOCCULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kw (7.5)

ITEM NO.	W14-RD-4110 B
SERVICE	HF 2'nd REDUCER DRIVER
Q'TY	1
TYPE	CYCLO REDUCER
CAPA.	m3/d (0.067)
MAT'L	STS304/STS304
POWER	kw (2.2)

ITEM NO.	W14-P-4110 B1/B2
SERVICE	HF 2'nd SLUDGE RETURN PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VVF), VORTEX
CAPA.	m3/hr.ea x mH 50 x 30 (0.29MPa)
MAT'L	SSC13/SSC19
POWER	kw (30)



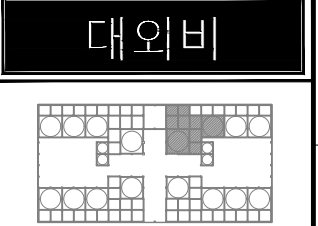
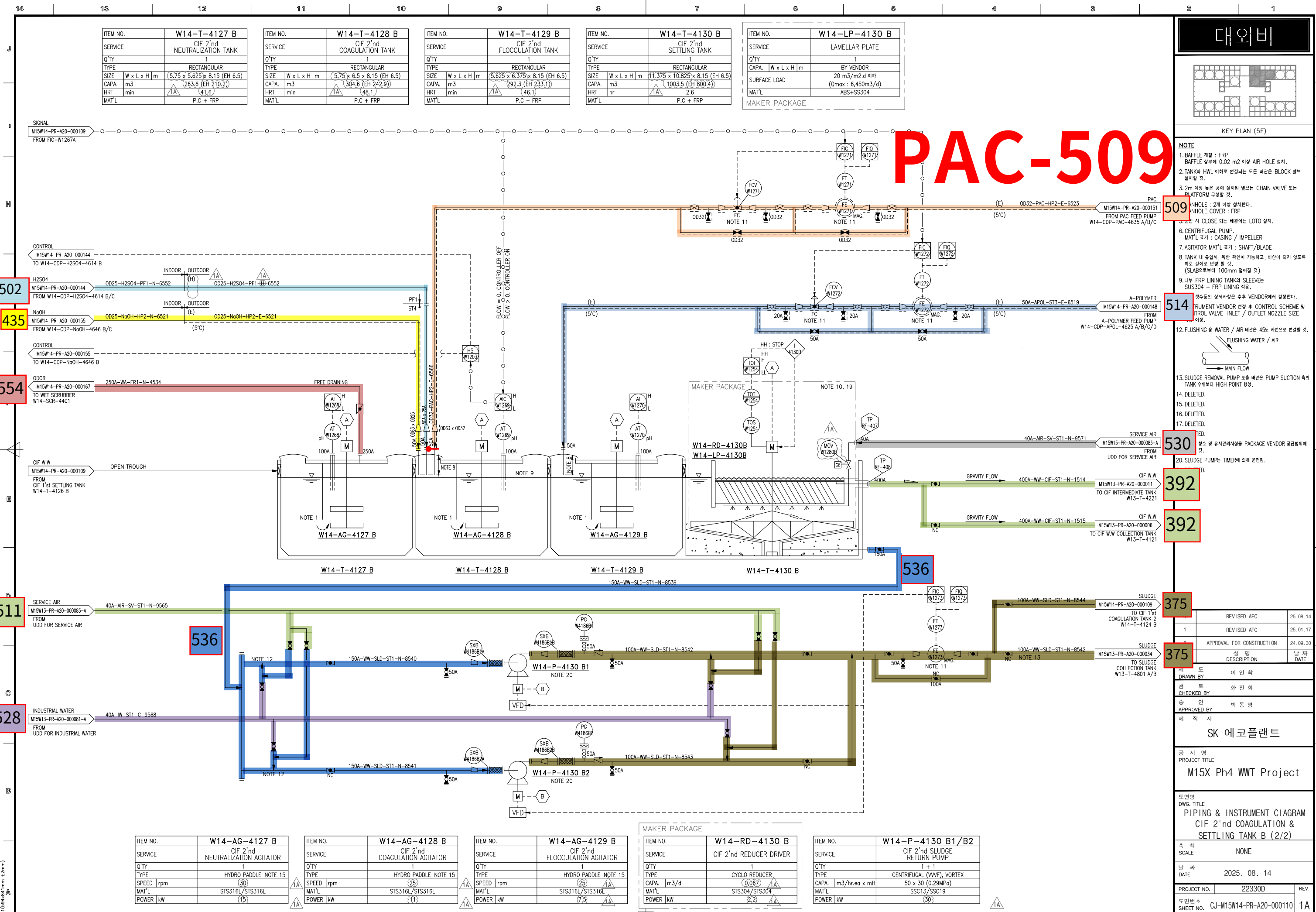


- NOTE
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
NK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브
설치 것.
 2. 3.2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는
PLATFORM 구성할 것.
 3. HOLE : 2개 이상 설치한다.
HOLE COVER : FRP
HOLE CLOSE 되는 배관에는 LOTO 설치.
 4. 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 5. 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 6. 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록
최소 길이로 변경 할 것.
B 으로부터 100mm 떨어질 것
 7. FRP LINING TANK의 SLEEVE는
504 + FRP LINING 적용.
 8. 10. Co(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것
VENDOR에서 결정한다.
라인은 상부에서 시수 주입하도록 설치한다.
 9. 14. W14-P-4121 A가 RUN 후 30초 후부터 동작한다.
 10. 15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및
CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
구간 투명 PVC 적용. (주입 여부 확인용)
 11. 17. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
 12. 18. SLUDGE REMOVAL PUMP를 배관은 PUMP SUCTION 속의
TANK 수위보다 HIGH POINT 형성.
 13. 19. SLUDGE PUMP는 TIMER에 의해 운전됨.
20. HIGH POINT에 SIPHON BREAKER 설치할것.
(CHECK VALVE의 경우 DUAL PLATE TYPE로 적용)
 14. 21. 소액의 공금 HEADER에서 주입되는 POINT 별
ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE,
SPEC. BREAK 까지 MIN. DISTANCE 적용하여,
적합 최소화 및 주입배관 고체를 용이하게 한다.
 15. 22. XVM SPEED CONTROLLER (ACCESSORY) 설치.
 16. 23. CONCRETE WALL OPENING을 통한 유체 이용
OPENING SIZE IS 800mm x 400mm

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE

제	도	인	학
DRAWN BY			
검	토	한	진
ED BY			
인	박	동	영
VED BY			
제	작	사	

공	사	명
PROJECT TITLE		
M15X Ph4 WWT Project		
도면명		
DWG. TITLE		
PIPING & INSTRUMENT CIAGRAM		
CIF 1'st COAGULATION &		
SETTLING TANK A (1/2)		
속	척	
SCALE	NONE	
날	짜	
DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호		
SHEET NO.	CJ-M15W14-PR-A20-000107	1A



- NOTE
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
 2. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. ANHOLE : 2개 이상 설치한다.
ANHOLE COVER : FRP
 5. LOTO 시 CLOSE 되는 배관에는 LOTO 설치.
 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 반영 할 것.
(SLAB으로부터 100mm 떨어진 것)
 9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 10. 각종의 상세사항은 추후 VENDOR에서 결정한다.
 11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 적용.
 12. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
 13. SLUDGE REMOVAL PUMP 또는 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 형성.
 14. DELETED.
 15. DELETED.
 16. DELETED.
 17. DELETED.
 18. DELETED.
 19. DELETED.
 20. SLUDGE PUMP는 TIMER에 의해 운전될 것.

REVISED AFC	25.08.14
REVISED AFC	25.01.17
APPROVAL FOR CONSTRUCTION	24.09.30
실명	날짜
DESCRIPTION	DATE

제 도 이 인 학
DRAWN BY
검 토 한 진 희
CHECKED BY
승 인 박 동 명
APPROVED BY
제 작 사

SK 에코플랜트
공 사 명
PROJECT TITLE
M15X Ph4 WWT Project

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
CIF 2'nd COAGULATION & SETTLING TANK B (2/2)
속 척
SCALE
NONE
날 짜
DATE
2025. 08. 14
PROJECT NO.
22330D
REV.
도면번호
SHEET NO.
CJ-M15W14-PR-A20-000110
1A

ITEM NO.	W14-T-4127 B
SERVICE	CIF 2'nd NEUTRALIZATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (5.75 x 5.625 x 8.15 (EH 6.5) (263.6 (EH 210.2))
CAPA.	m3 (41.6)
HRT	min (41.6)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4128 B
SERVICE	CIF 2'nd COAGULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (5.75 x 6.5 x 8.15 (EH 6.5) (304.6 (EH 242.9))
CAPA.	m3 (48.1)
HRT	min (48.1)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4129 B
SERVICE	CIF 2'nd FLOCCULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (5.625 x 6.375 x 8.15 (EH 6.5) (292.3 (EH 233.1))
CAPA.	m3 (46.1)
HRT	min (46.1)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4130 B
SERVICE	CIF 2'nd SETTLING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (11.375 x 10.825 x 8.15 (EH 6.5) (1003.5 (EH 800.4))
CAPA.	m3 (2.6)
HRT	hr (2.6)
MAT'L	P.C + FRP

ITEM NO.	W14-LP-4130 B
SERVICE	LAMELLAR PLATE
Q'TY	1
CAPA.	W x L x H m BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 6,450m3/d)
MAT'L	ABS+SS304

ITEM NO.	W14-AG-4127 B
SERVICE	CIF 2'nd NEUTRALIZATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (30)
MAT'L	STS316L/STS316L
POWER	kw (15)

ITEM NO.	W14-AG-4128 B
SERVICE	CIF 2'nd COAGULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kw (11)

ITEM NO.	W14-AG-4129 B
SERVICE	CIF 2'nd FLOCCULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kw (7.5)

ITEM NO.	W14-RD-4130 B
SERVICE	CIF 2'nd REDUCER DRIVER
Q'TY	1
TYPE	CYCLO REDUCER
CAPA.	m3/d (0.067)
MAT'L	STS304/STS304
POWER	kw (2.2)

ITEM NO.	W14-P-4130 B1/B2
SERVICE	CIF 2'nd SLUDGE RETURN PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (WVF), VORTEX
CAPA.	m3/hr.ea x mH 50 x 30 (0.29MPa)
MAT'L	SSC13/SSC19
POWER	kw (30)

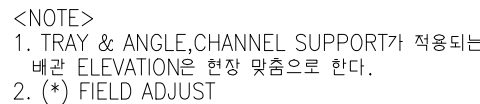
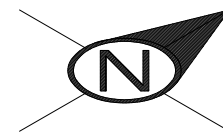


CLIENT  ①

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
PAC		22330D	
LINE NO.	OD90-PAC-HP2-N-6513	SH'T NO.	REV.
		01 / 02	1

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	—	DESIGN	6.4	60	UNIT 4FA						
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동
						REF. P & ID CJ-M15W14-PR-A20-000151		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP	

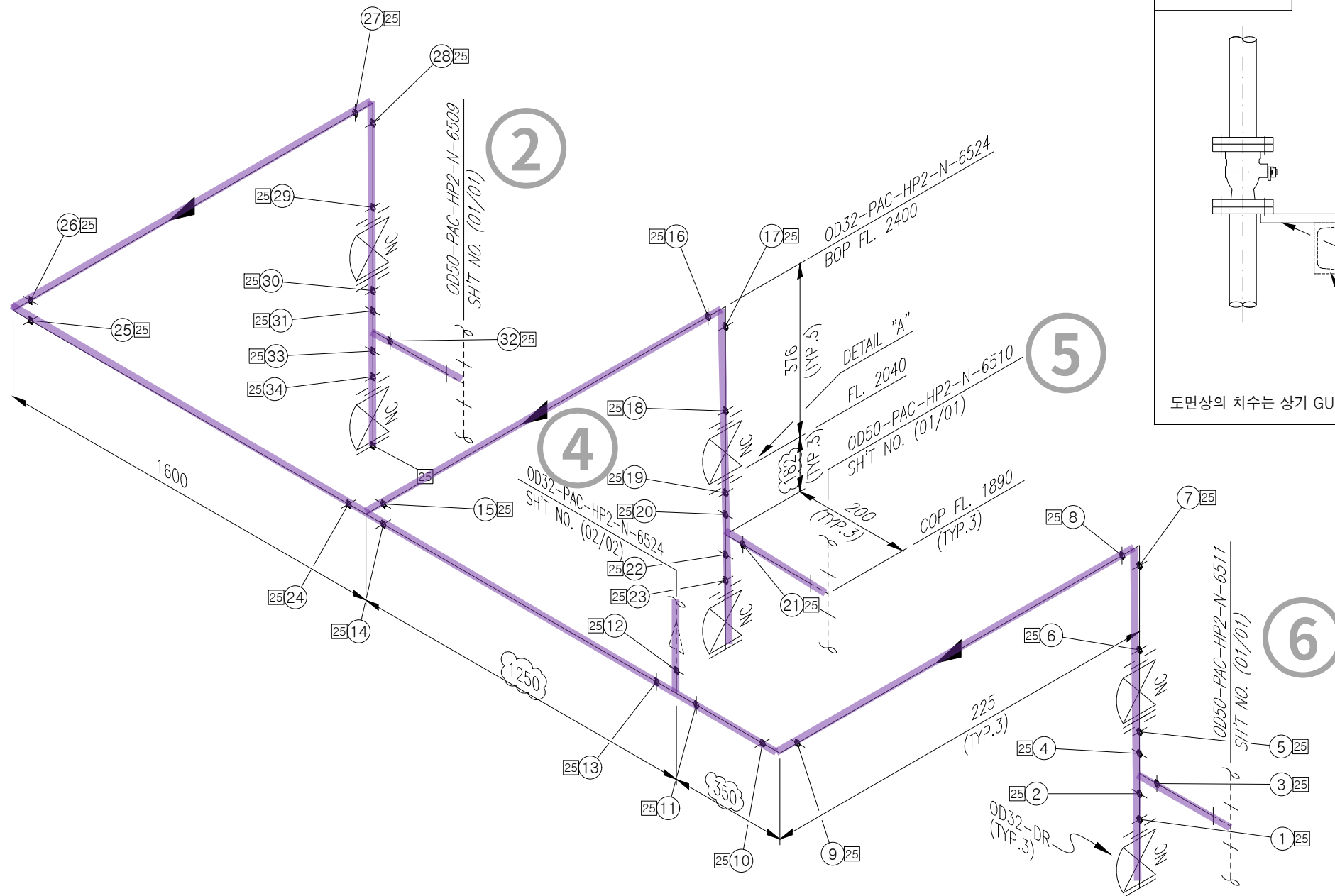


CLIENT  ②

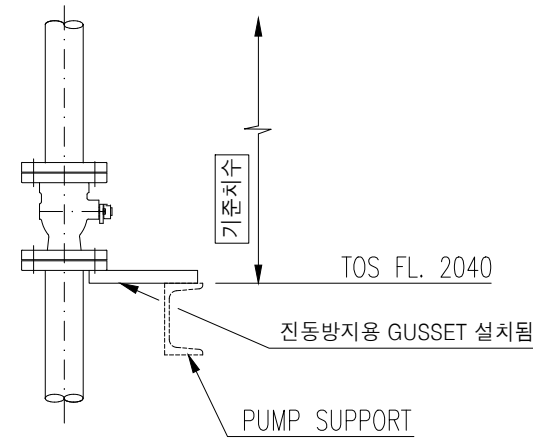
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. ("C)	AREA W14							CONTRACTOR 	PROJECT NAME
	THICK.	-	DESIGN	6.4	60	UNIT 4FA								M15X Ph4 Project
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A	 1	25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY PAC	JOB NO.
						REF. P & ID CJ-M15W14-PR-A20-000151	 0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영		22330D
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO. OD50-PAC-HP2-N-6509	SH'T NO. 01 / 01 REV. 1



ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD32	6.4M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
E90	OD32	5EA	ELBOW 90 DEG, HDPE PE100, PE, SOCKET FUSION,
			PN16/SDR11
T	OD32	5EA	TEE EQUAL, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
VD	OD32	6EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-PP /
			TRIM-EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL
F	OD32	9EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
			LAP JOINT-STS304 / STUB END-HDPE PE100
FB	OD32	3EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STS304
			/ LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD32	9EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	3EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD32	12SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	OD32	12EA	SHIELD
LOTO	OD32	6EA	LOTO



DETAIL "A"



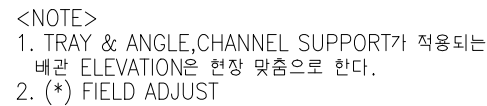
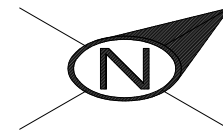
도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

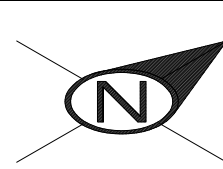
* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1	60	UNIT 4FA							M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-04008-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000151		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	PAC	22330D	
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
													0D32-PAC-HP2-N-6524	01 / 02	1



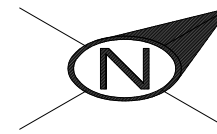
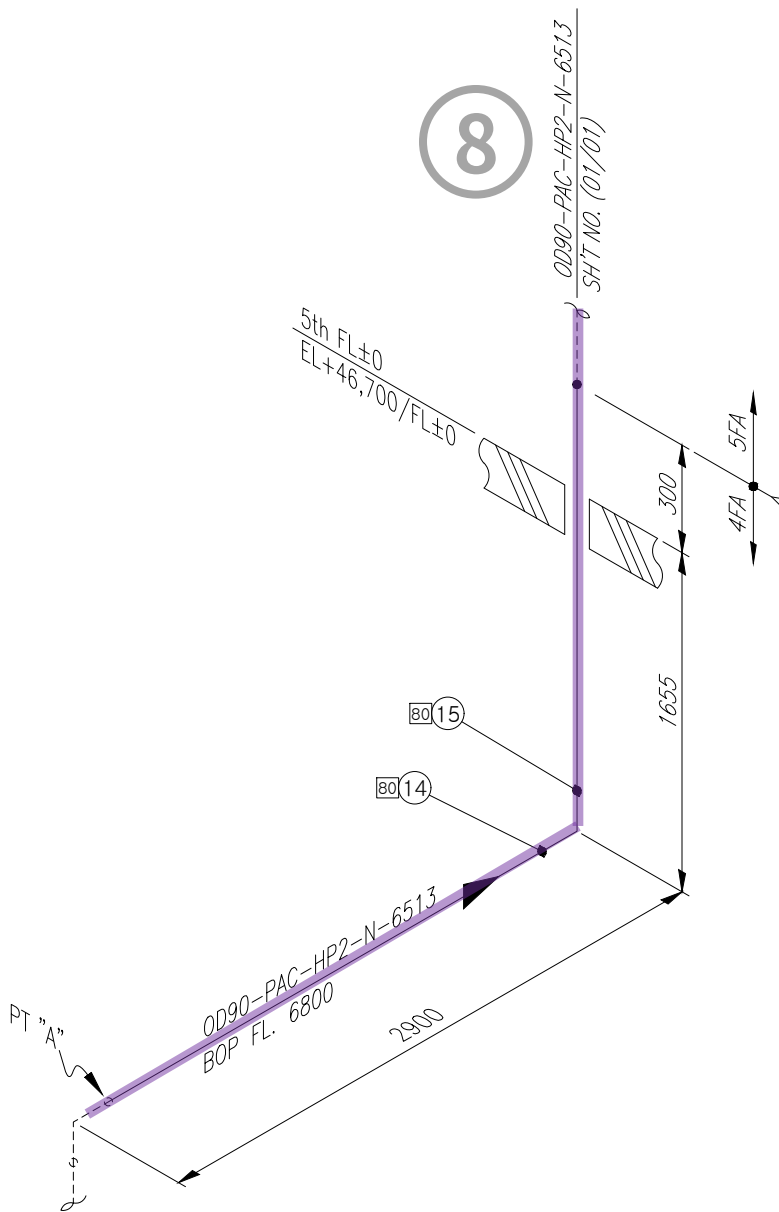
CLIENT  

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR 	PROJECT NAME	
	THICK.	–	DESIGN	6.4	60	UNIT 4FA								M15X Ph4 Project	
			OPERATING	4.7	AMB	REF. PLAN DWG.		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	PAC	JOB NO.
						CJ-M15W14-PI-A30-040013-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			22330D
						REF. P & ID CJ-M15W14-PR-A20-000151	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	0D50-PAC-HP2-N-6510	SH'T NO. 01 / 01



CLIENT		 	
CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY PAC		JOB NO. 22330D	
LINE NO. OD50-PAC-HP2-N-6511		SH'T NO. 01 / 01	REV. 1

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RC51 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1 * STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1													CLIENT  			
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	6.4	60	UNIT	4FA							M15X Ph4 Project		
			OPERATING	4.7	AMB	REF. PLAN DWG.			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040013-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	PAC	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000151								OD50-PAC-HP2-N-6511	01 / 01	1



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

[illegible]

CLIENT		 	
CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY PAC		JOB NO. 22330D	
LINE NO.	OD90-PAC-HP2-N-6513	SH'T NO. 02 / 02	REV. 0

[illegible]

<NOTE>

1. (*) FIELD ADJUST
2. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT



8

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	6.3	60	UNIT 5FA								M15X Ph4 Project	
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050005-A							COMMODITY	JOB NO.	
								25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영		22330D	
						REF. P & ID CJ-M15W14-PR-A20-000151	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
													OD90-PAC-HP2-N-6513	01 / 01	0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출을 할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	6.3	60	UNIT RFA						
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W13-PI-A30-R10005-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영
								25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
						REF. P & ID CJ-M15W14-PR-A20-000151	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'N

CLIENT		 	
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY	PAC	JOB NO.	
		22330D	
LINE NO.	OD90-PAC-HP2-E-6513	SH'T NO.	REV.
		01 / 04	1



<NOTE>
1. (*) FIELD ADJUST

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있습니다.



1. (*) FIELD ADJUST

CLIENT



11

CONTRACTOR

PROJECT NAME	M15X Ph4 Project
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COMMODITY	PAC
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LINE NO.	0032-PAC-HP2-F-6517
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JOB NO.	
22330D	
SH'T NO.	REV.
01 / 02	1



CONTRACTOR 		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY	PAC	JOB NO.	
		22330D	
LINE NO.	OD32-PAC-HP2-E-6517	SH'T NO. 02 / 02	REV. 1

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유도자의 불법 사용으로 피해가 발생할 경우 유도자에게 민/형사상 책임을 요구할 수 있음



CLIENT  



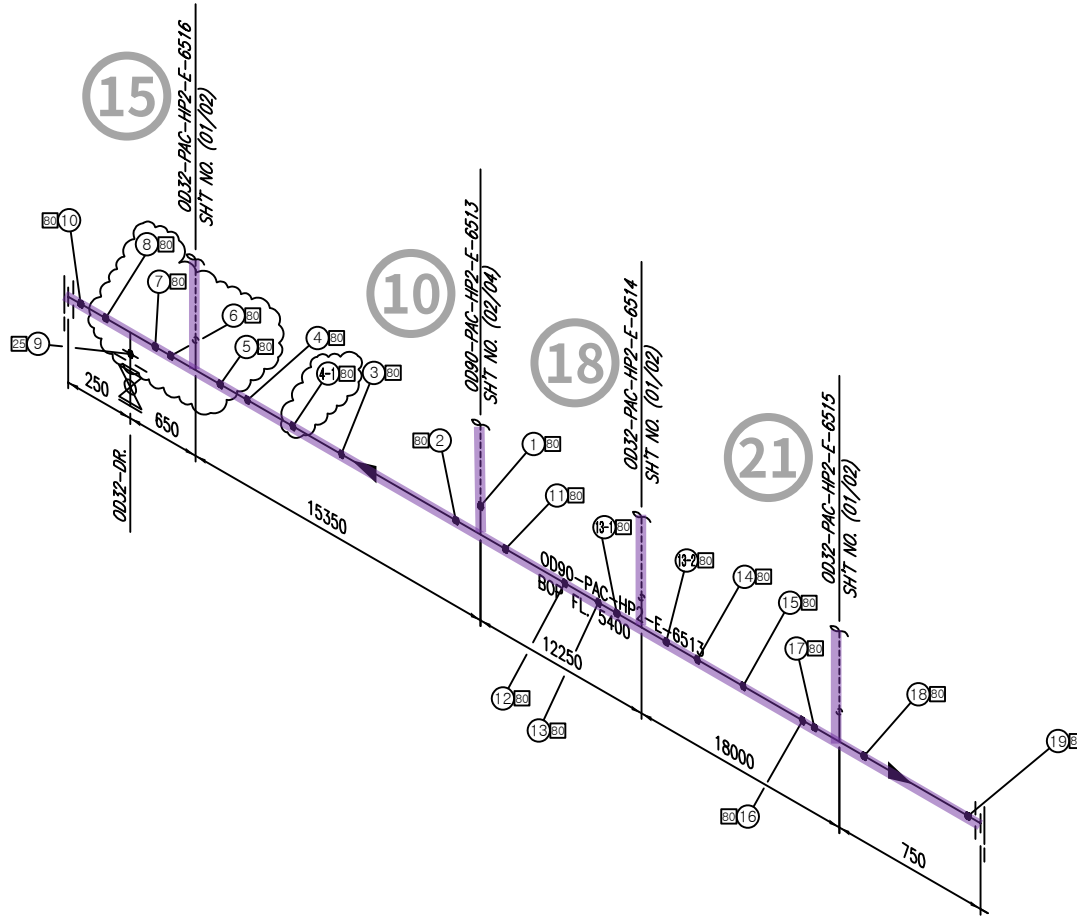
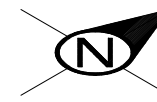
COMMODITY	PAC	JOB NO. 22330D
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LINE NO.	OD32-PAC-HP2-E-6558	SH'T NO.	REV.
		01 / 01	1



INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	6.3	60	UNIT RFA						
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영
						REF. P & ID CJ-M15W14-PR-A20-000105		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유품자의 불법 사용으로 피해가 발생할 경우 유품자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * SUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* SUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB2L * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD1 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14
	THICK.	25	DESIGN	6.3	60	UNIT	RFA
			OPERATING	4.7	AMB	REF. PLAN DWG.	
						CJ-M15W14-PI-A30-R10008-B	
						REF. P & ID	
						CJ-M15W14-PR-A20-000151	



25.03.31

REVISED AS MARKED

황래중 안남후 박동영



25.01.24

APPROVED FOR CONSTRUCTION

황래중 안남후 박동영

REV. NO.

DATE

DESCRIPTION

BY

CHK

APP'D

CLIENT



14

CONTRACTOR	PROJECT NAME
SK ecoplant	M15X PH4 Project
COMMODITY	JOB NO.
PAC	22330D
LINE NO.	SH'T NO. REV.
OD90-PAC-HP2-E-6513	04 / 04 1

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.

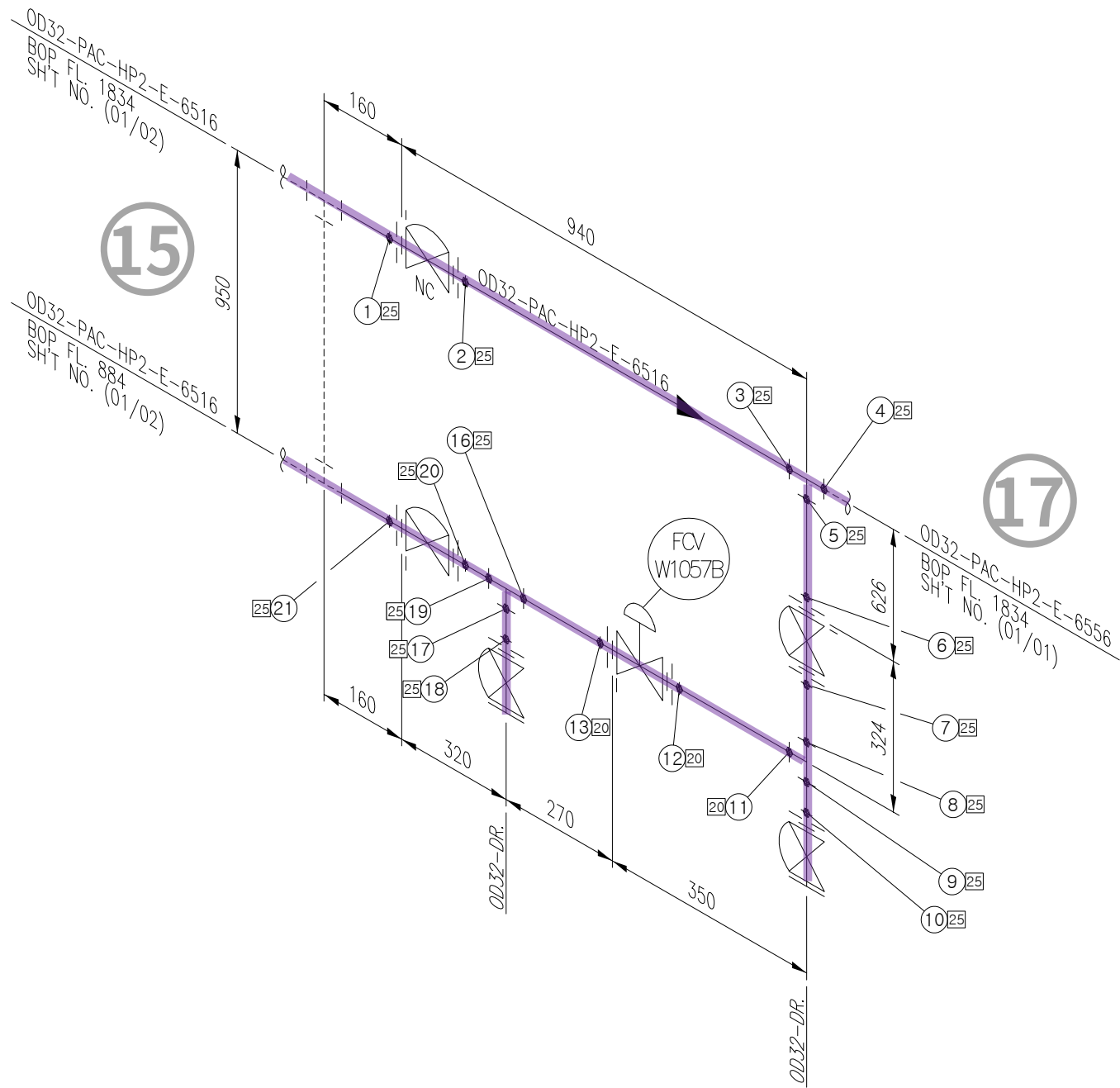
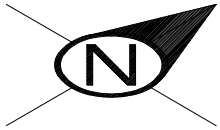


TOR  SK ecoplant



INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	6.3	60	UNIT RFA						
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10009-B		25.03.31	REVISED AS MARKED	황래중	안남후	박동영
						REF. P & ID CJ-M15W14-PR-A20-000104		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP	

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. (*) FIELD ADJUST



ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD32	3.2M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD25	0.6M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD32	2EA	TEE EQUAL, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
TR	OD32xOD25	1EA	TEE RED, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
RC	OD32xOD25	1EA	REDUCER CON, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
VD	OD32	5EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL
F	OD32	10EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
F	OD25	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
FB	OD32	2EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STS304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD32	10EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	2EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD32	12SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	OD25	2SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
LOTO	OD32	1EA	LOTO
INC	25tXOD32	3.2M	관 보온재(고무발포) + 알루미늄 CASING
INC	25tXOD25	0.6M	관 보온재(고무발포) + 알루미늄 CASING

CLIENT



CONTRACTOR	PROJECT NAME
SK ecoplant	M15X Ph4 Project
COMMODITY	JOB NO.
PAC	22330D
LINE NO.	SH'T NO. REV.
OD32-PAC-HP2-E-6516	02 / 02 1

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14										
	THICK.	25	DESIGN	6.3	60	UNIT	RFA										
			OPERATING	4.7	AMB	REF. PLAN DWG.											
						CJ-M15W14-PI-A30-R10009-A											
						REF. P & ID											
						CJ-M15W14-PR-A20-000104											

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.



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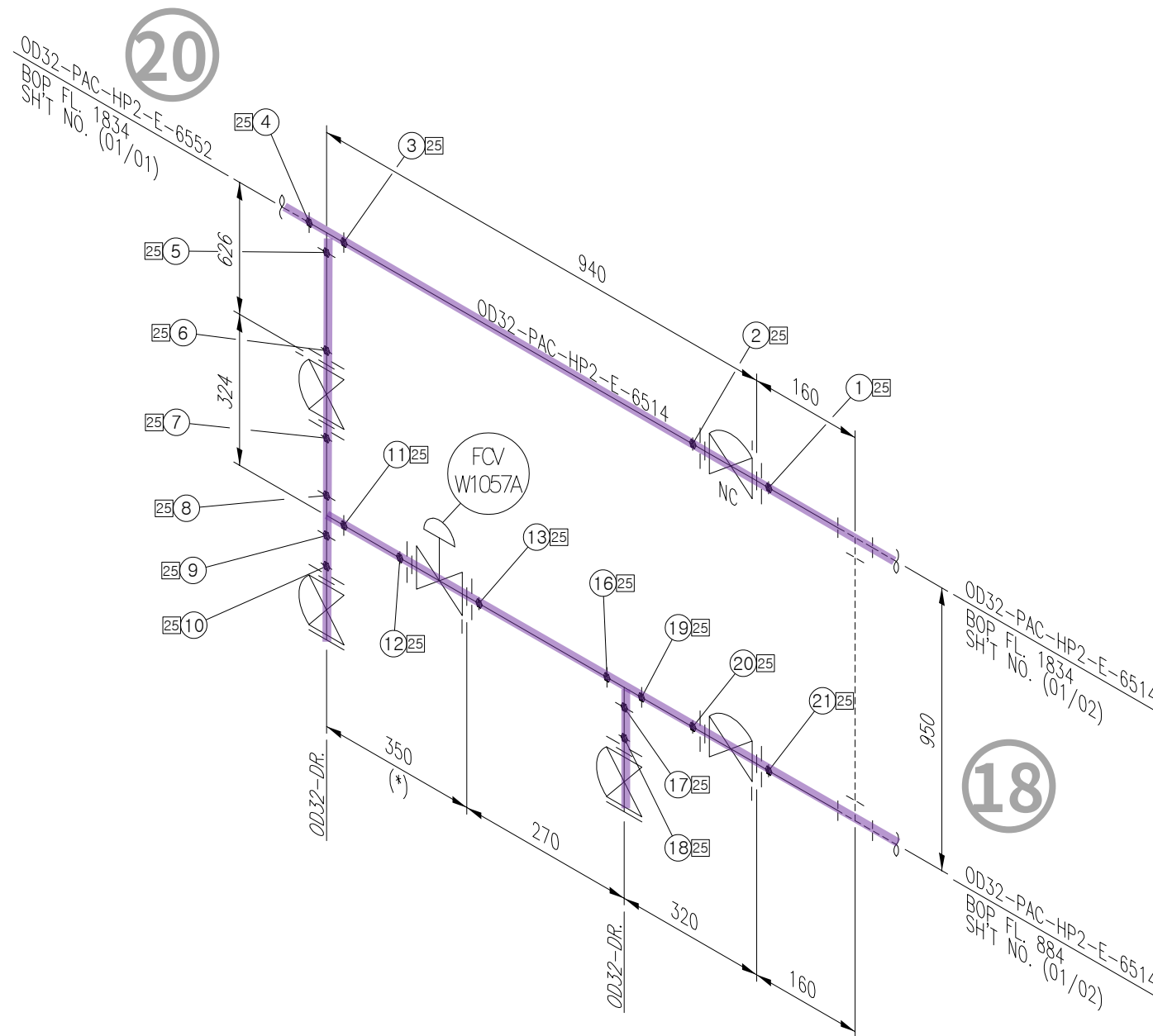
<NOTE>
1. (*) FIELD ADJUST

CONTRACTOR 

CONTRACTOR 



ITEM	SIZE	Q'TY	DESCRIPTION
		HP2	
P	OD32	3.2M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD25	0.6M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD32	2EA	TEE EQUAL, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
TR	OD32xOD25	1EA	TEE RED, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
RC	OD32xOD25	1EA	REDUCER CON, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
VD	OD32	5EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL
F	OD32	10EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
F	OD25	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
FB	OD32	2EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STS304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD32	10EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	2EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD32	12SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	OD25	2SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
LOTO	OD32	1EA	LOTO
INC	25tXOD32	3.2M	관 보온재(고무발포) + 알루미늄 CASING
INC	25tXOD25	0.6M	관 보온재(고무발포) + 알루미늄 CASING



<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB10/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	6.3	60	UNIT RFA								M15X Ph4 Project	
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	PAC	JOB NO.
								25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			22330D
						REF. P & ID CJ-M15W14-PR-A20-000102	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	0D32-PAC-HP2-E-6514	SH'T NO. 02 / 02

경고 : 본 법원은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 독도복제 및 사용, 복제, 제3자에게 공개, 누설, 전출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 물을 수 있다.



1. (*) FIELD ADJUST

CLIENT



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CONTRACTOR 

PROJECT NAME
M15X Ph4 Project

COMMODITY	PAC
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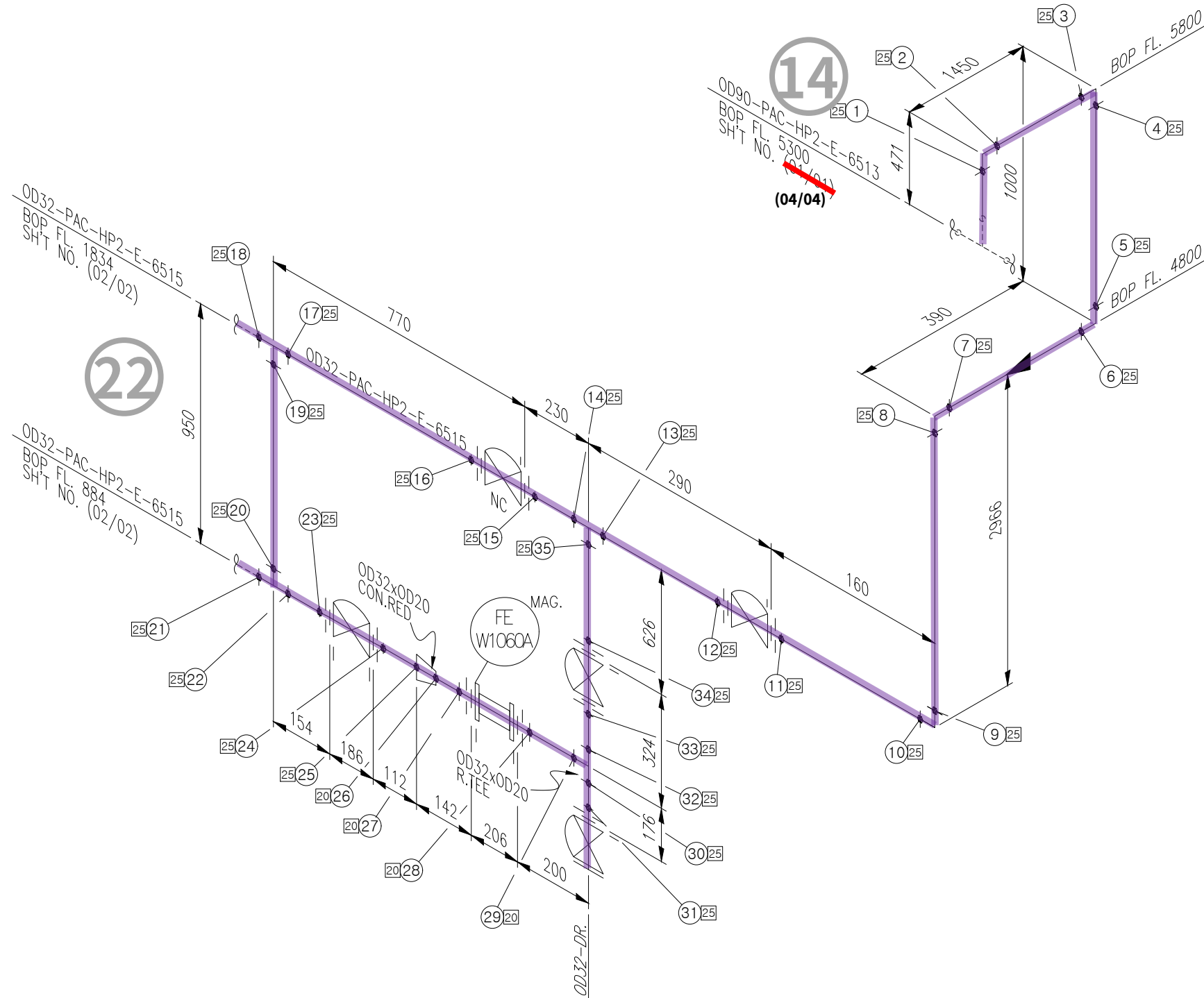
JOB NO.
22330D

LINE NO.	0D32-PAC-HP2-E-6552
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SH'T NO. 01 / 01	REV. 1
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ITEM	SIZE	Q'TY	DESCRIPTION
		HP2	
P	OD32	10.3M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD20	0.6M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
E90	OD32	5EA	ELBOW 90 DEG, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD32	3EA	TEE EQUAL, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
TR	OD32xOD20	1EA	TEE RED, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
RC	OD32xOD20	1EA	REDUCER CON, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
VD	OD32	5EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL
F	OD32	9EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
F	OD20	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
FB	OD32	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STS304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD32	9EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD20	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD32	10SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	OD20	2SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
LOTO	OD32	1EA	LOTO
INC	25tXOD32	10.3M	관 보온재(고무발포) + 알루미늄 CASING
INC	25tXOD20	0.6M	관 보온재(고무발포) + 알루미늄 CASING



<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME		
	THICK.	25	DESIGN	6.3	60	UNIT	RFA							M15X Ph4 Project		
			OPERATING	4.7	AMB	REF. PLAN DWG.			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-R10008-B			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	PAC	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000103								0D32-PAC-HP2-E-6515	01 / 02	0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



CONTRACTOR 

COMMODITY	PAC
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LINE NO.	OD32-PAC-HP2-E-6554
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SH'T NO. 01 / 01	REV. 1
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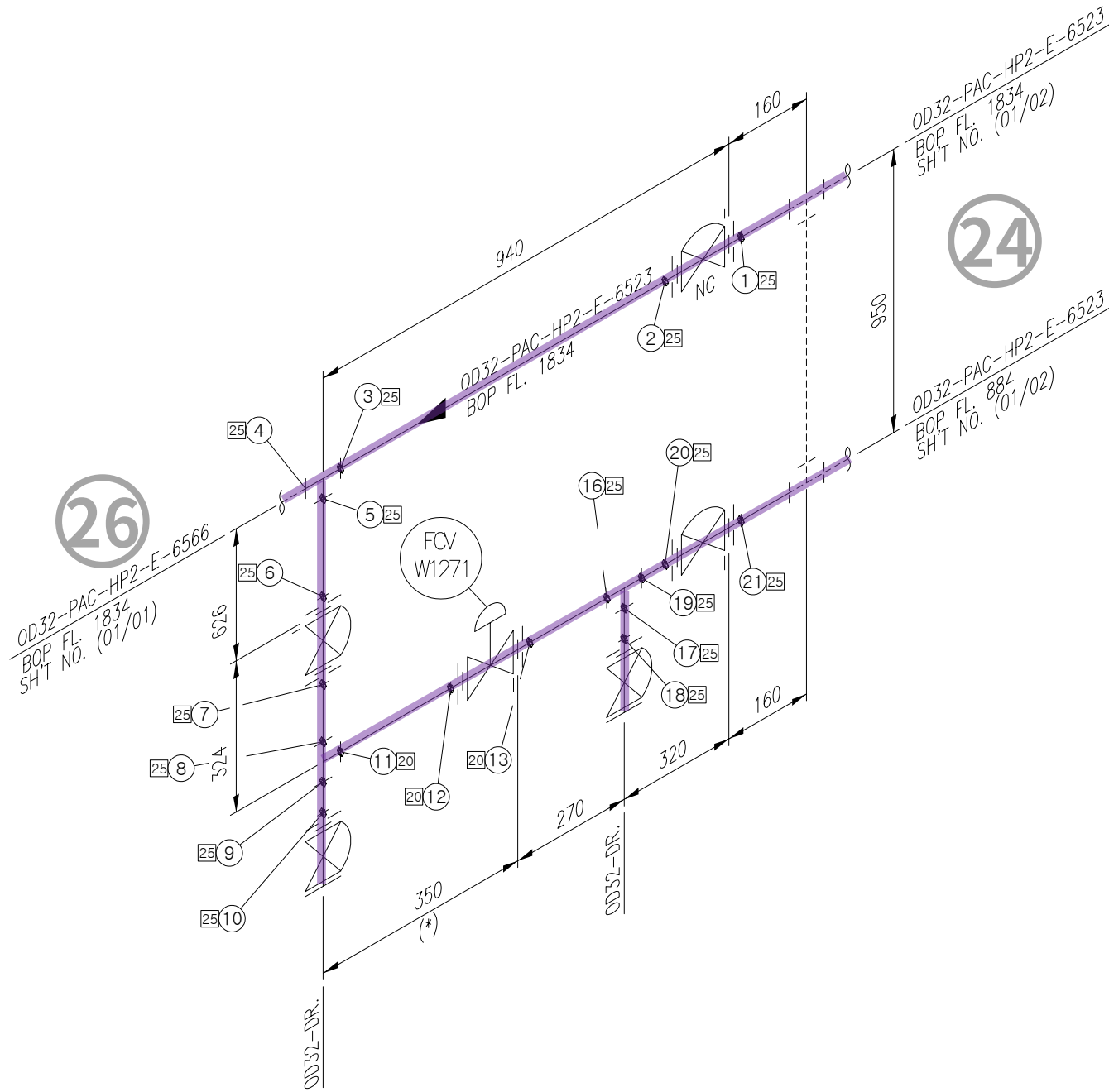
1. (*) FIELD ADJUST

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경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



ITEM	SIZE	Q'TY	DESCRIPTION
		HP2	
P	OD32	3.2M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD25	0.6M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD32	2EA	TEE EQUAL, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
TR	OD32xOD25	1EA	TEE RED, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
RC	OD32xOD25	1EA	REDUCER CON, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
VD	OD32	5EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL
F	OD32	10EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
F	OD25	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
FB	OD32	2EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STS304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD32	10EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	2EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD32	12SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	OD25	2SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
LOTO	OD32	1EA	LOTO
INC	25tXOD32	3.2M	관 보존재(교무벌포) + 알루미늄 CASING
INC	25tXOD25	0.6M	관 보존재(교무벌포) + 알루미늄 CASING



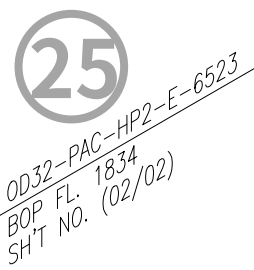
<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1O/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	25	DESIGN	6.3	60	UNIT RFA								M15X Ph4 Project		
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10002-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	PAC	JOB NO.	
								23.02.28	APPROVED FOR CONSTRUCTION	정경만	이광연	박동영			22330D	
						REF. P & ID CJ-M15W14-PR-A20-000110	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	0D32-PAC-HP2-E-6523	SH'T NO. 02 / 02	REV. 1

경고 : 피고인 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 독점적 외 사용, 복제, 제3자에게 공개, 누설, 정보출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 물을 수 있다.



FULL COUPLING : FC * HALF COUPLING : HF/HF1
VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
SPN * EMERGENCY SHOWER : SSH
H * SPECIAL BOLT & NUT : BL/BL1

CLIENT  26

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	6.3	60	UNIT RFA						
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10002-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영
						REF. P & ID CJ-M15W14-PR-A20-000110		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP

CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY PAC		JOB NO. 22330D	
LINE NO. OD32-PAC-HP2-E-6566		SH'T NO. 01 / 01	REV. 1



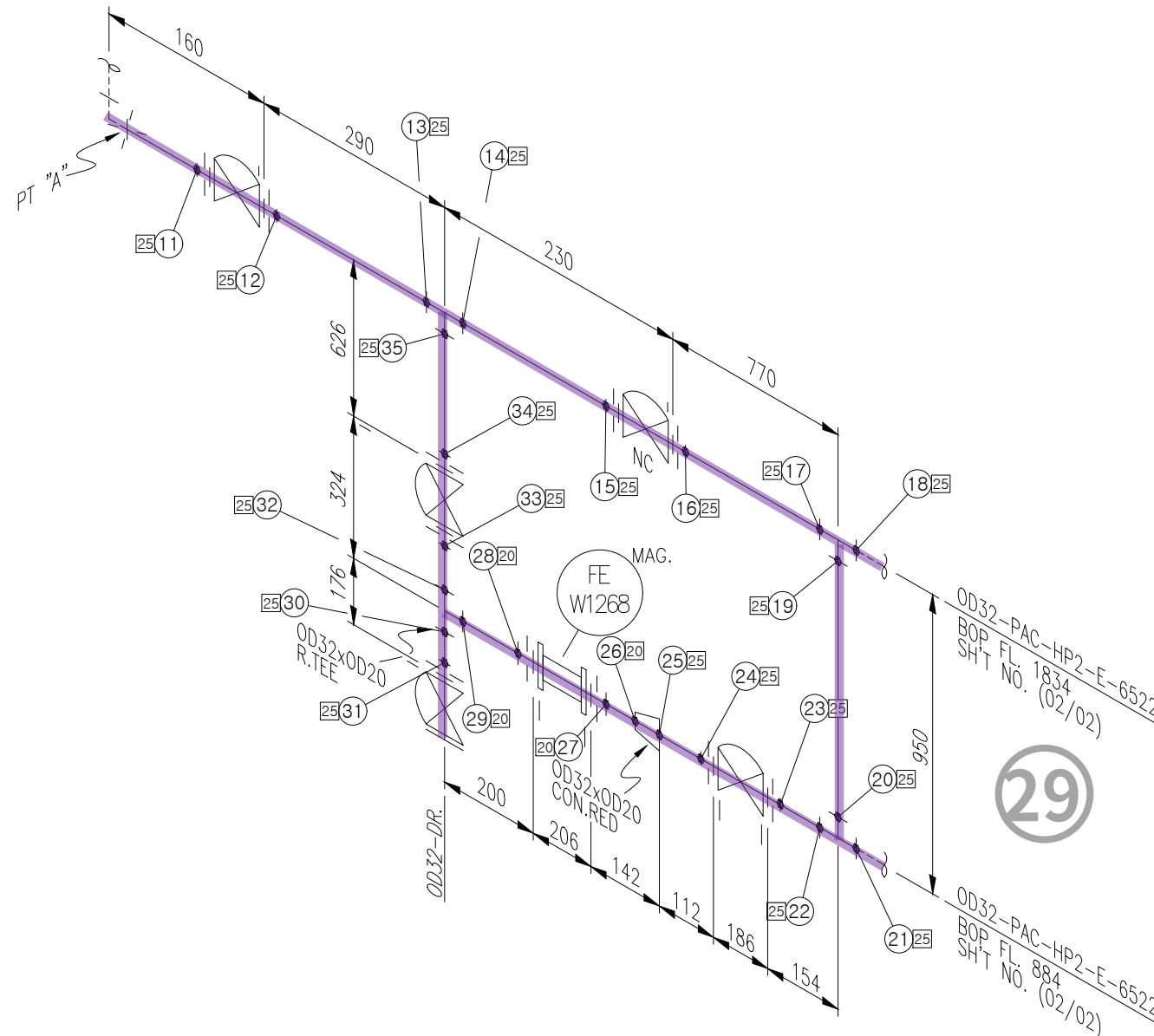
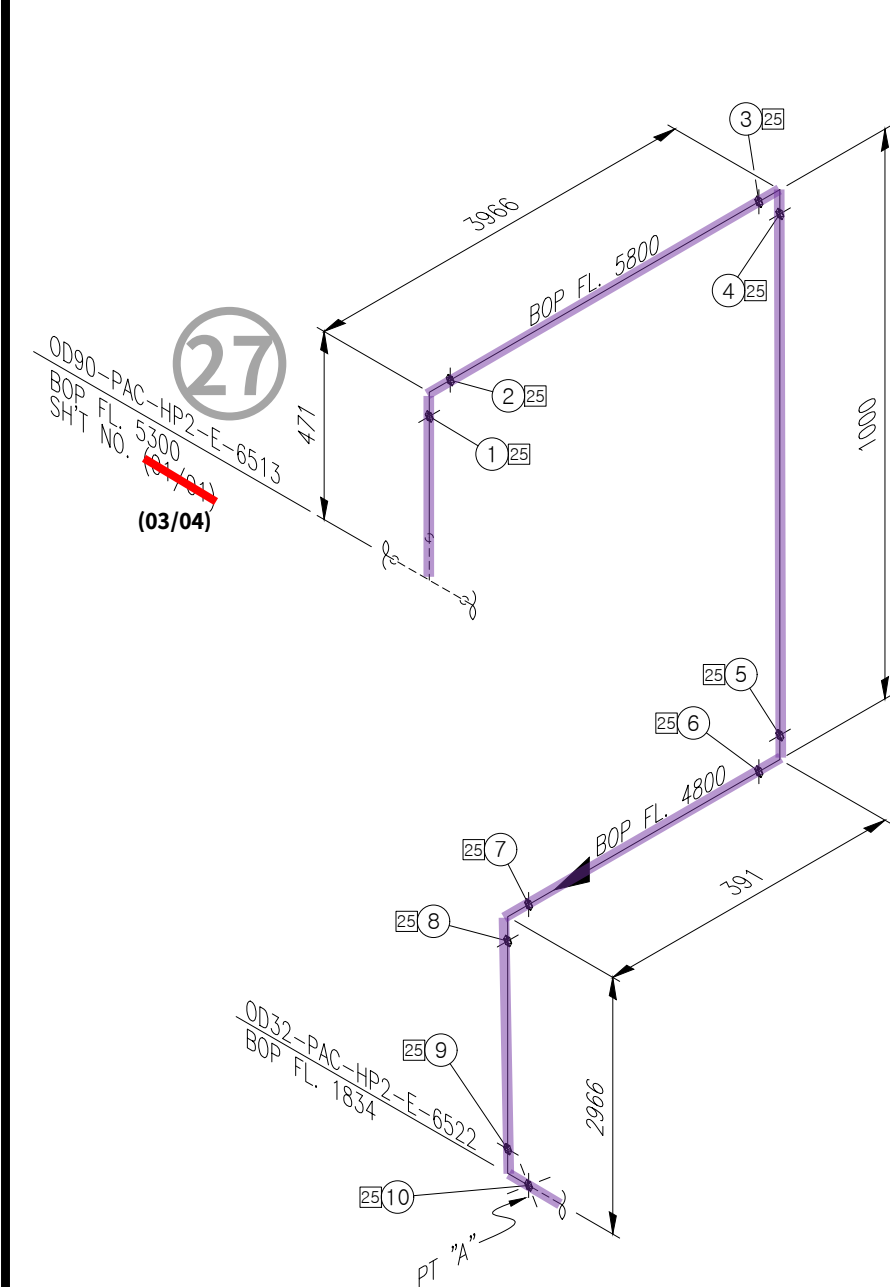
CLIENT  



INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	6.3	60	UNIT RFA						
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10002-B		25.03.31	REVISED AS MARKED	황래중	안남후	박동영
						REF. P & ID CJ-M15W14-PR-A20-000151		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D

[illegible]

CLIENT		 	
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY	PAC	JOB NO.	
		22330D	
LINE NO.	OD32-PAC-HP2-E-6522	SH'T NO.	REV.
		01 / 02	1



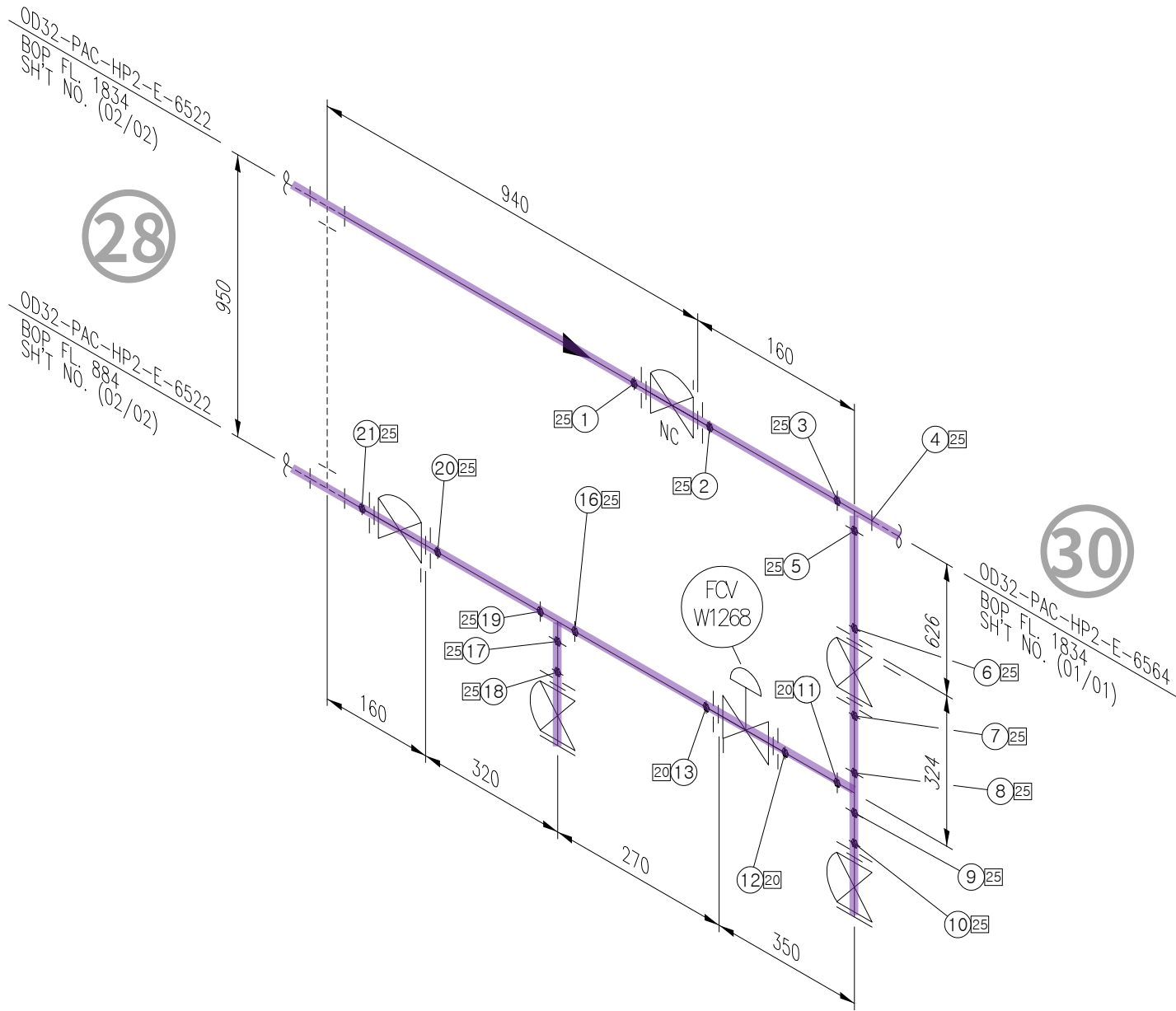
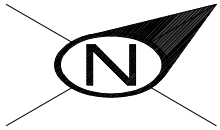
<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VBLO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	6.3	60	UNIT	RFA								M15X Ph4 Project	
			OPERATING	4.7	AMB	REF. PLAN DWG.			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-R10003-B			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	PAC	22330D	
						REF. P & ID CJ-M15W14-PR-A20-000109		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	0D32-PAC-HP2-E-6522	SH'T NO. 01 / 02

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출을 할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	6.3	60	UNIT RFA								M15X Ph4 Project	
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10003-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY PAC	JOB NO. 22330D	
						REF. P & ID CJ-M15W14-PR-A20-000109		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO. OD32-PAC-HP2-E-6522	SH'T NO. 02 / 02	REV. 1
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD32	3.2M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD25	0.6M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD32	2EA	TEE EQUAL, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
TR	OD32xOD25	1EA	TEE RED, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
RC	OD32xOD25	1EA	REDUCER CON, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
VD	OD32	5EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL
F	OD32	10EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
F	OD25	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
FB	OD32	2EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STS304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD32	10EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	2EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD32	12SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	OD25	2SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
LOTO	OD32	1EA	LOTO
INC	25tXOD32	3.2M	관 보온재(고무발포) + 알루미늄 CASING
INC	25tXOD25	0.6M	관 보온재(고무발포) + 알루미늄 CASING

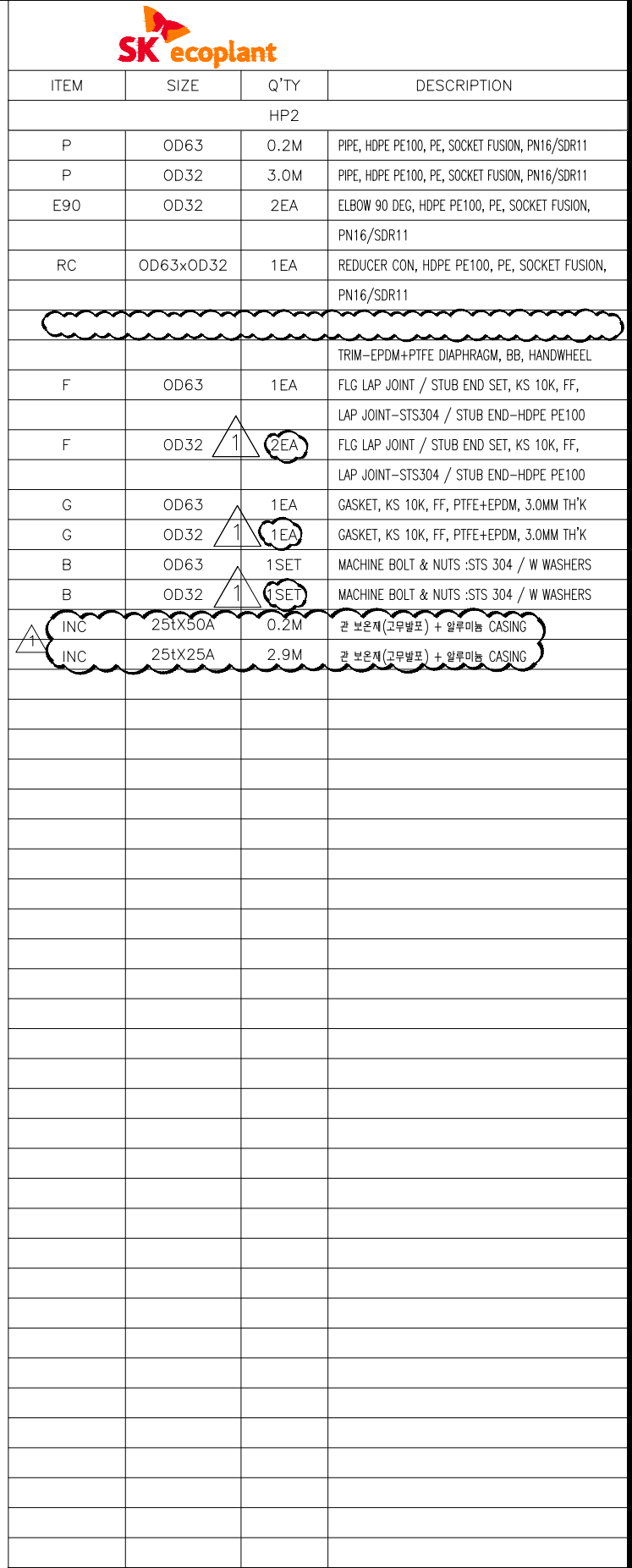
CLIENT





CLIENT		 	
CONTRACTOR 		PROJECT NAME	
		M15X Ph4 Project	
명	COMMODITY	JOB NO.	
	PAC	22330D	
명	LINE NO.	SH'T NO.	REV.
D	0D32-PAC-HP2-E-6520	02 / 02	1

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	25	DESIGN	6.3	60	UNIT RFA								M15X Ph4 Project		
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10002-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	PAC	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000107		23.02.28	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	OD32-PAC-HP2-E-6560	SH'T NO. 01 / 01	REV. 1
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				



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TOR  SK ecoplant

TY	PAC
	OD32-PAC-HP2-E-



FULL COUPLING : FC * HALF COUPLING : HF/HF1
VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
SPN * EMERGENCY SHOWER : SSH
SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT



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TOR  SK ecoplant